

Models of Higher Brain Functions

Causal inference in multisensory perception

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1 Introduction

Multisensory perception plays an important role in how humans perceive and interpret the world around them. When presented with stimuli from multiple sensory modalities, such as visual and auditory stimuli, the brain integrates these cues to come up with a causal inference. This process can be modeled using Bayesian statistics, to demonstrate how a human can optimally infer stimuli location and cause from the evidence available to them. To understand the mechanisms behind this process, we investigated and analyzed a statistical model of causal inference for a multisensory perception task, based on the work by K rting et al. (2007).

The model aims to explain the behavior of subjects in an experimental setup where they are asked to identify the locations of both auditory and visual stimuli by pressing one button from a set of 5 buttons for each type of stimulus. The model at hand explains and models the process of causal inference and integration in multisensory perception as an optimal Bayesian observer that not only infers the original stimuli locations from two sensory signals (visual, s_V , and auditory, s_A) but also whether these signals come from a common cause (C). The inference process is done by the brain, which lacks access to the exact stimuli locations but instead relies on noisy measurements (visual, x_V , and auditory, x_A). Despite this uncertainty in perception, the system should accurately estimate the stimuli locations ($\hat{S}_V$ and $\hat{S}_A$). In addition, this previous task is further complicated since the

system also infers whether the stimuli come from a common cause (C), which also affects the estimated locations of stimuli ($\hat{S}_V$ and $\hat{S}_A$).

To understand the relationships between the variables and parameters of this Bayesian model, we implemented and analyzed the model by varying parameter settings.

additionally, the model was used with a default set of parameters to generate simulated "experimental" data that should resemble the data acquired from real subjects in this experimental setting.

Subsequently, two different approaches were investigated to fit the model to the generated experimental data, i.e., estimate the default parameters that generated the data. Namely, a brute-force parameter space search and Markov Chain Monte Carlo (MCMC) sampling.

In both methods, maximum log-likelihood estimation was used to evaluate the performance of the parameter combination being tested.

2 Materials and Methods

2.1 Model Parameters

The causal inference model depends on five parameters that characterize the subject's perception of the environment and priors about it, uncertainties in the sensory systems involved as well as the human bias to estimate that stimuli come from central locations. the model parameters are:

- p_{common} : The prior probability of a single cause versus two causes.
- σ_v : Uncertainty of visual perception.
- σ_a : Uncertainty of auditory perception.
- μ_p : mean location arising from human bias to infer stimuli come from certain locations.
- σ_p : Spatial variability of the aforementioned human bias.

2.2 Experimental Data Generation

To generate the synthetic experimental data, we used the causal inference Model to simulate the behavior of subjects in this experimental setting. 10

thousand data points were generated for each simulated subject ($N = 10000$). Here is the step-by-step process:

1. **Parameter initialization:** We set the values of the model parameters for this study to be ($p_{common} = 0.8$, $\sigma_v = 0.6$, $\sigma_a = 3.1$, $\sigma_p = 15$, and $\mu_p = 0$).

2. **Stimulus Pair Generation:** We generated pairs of visual and auditory stimuli (V and A) from permutations of a set of five predefined values that correspond to the five possible stimulus presentation locations.

3. **Noise Addition** To mimic human sensory perception, we added noise to the generated stimuli locations. the noise was assumed to come from normal distributions with standard deviations σ_v and σ_a for vision and audition, respectively.

4. **Common Cause Inference:** Based on the proximity of the stimuli locations, we calculated the posterior probability of a common cause ($C = 1$) using the Model’s likelihood function adapted from K rting et al:

$$p(C = 1 \mid x_V, x_A) = \frac{p(x_V, x_A \mid C = 1) p_{common}}{p(x_V, x_A \mid C = 1) p_{common} + p(x_V, x_A \mid C = 2) (1 - p_{common})} \quad (1)$$

5. **Location Estimation:** Using the calculated posterior probability of a common cause and other variables in the model, the estimated locations of visual and auditory stimuli for each stimulus pair are calculated using the following equations adapted from K rting et al:

$$\hat{s}_V = p(C = 1 \mid x_V, x_A) \hat{s}_{V,C=1} + (1 - p(C = 1 \mid x_V, x_A)) \hat{s}_{V,C=2} \quad (2)$$

$$\hat{s}_A = p(C = 1 \mid x_V, x_A) \hat{s}_{A,C=1} + (1 - p(C = 1 \mid x_V, x_A)) \hat{s}_{A,C=2} \quad (3)$$

6. **Quantization:** To simulate the recorded responses of subjects in the actual experiment, we assigned each of the estimated locations in our simulation $\hat{s}_V$ and $\hat{s}_A$ to one of five bins corresponding to the button presses the subjects made in the actual experiment (i.e., left, slightly left, centered, slightly right, right).

2.3 Model Fitting and Evaluation

Two different methods were used to fit the model to the data. the first method involves a brute-force parameter space search. Particularly, we varied the model’s parameters within reasonable pre-defined ranges for each parameter (particularly: $0 \leq p_{common} \leq 1, 0 < \sigma_{v,a,p} < 20$), this resulted in a total of 10 thousand parameter combinations and the fit of each of these combinations was evaluated to determine the combination that best fits the experimental data.

The second method utilized for fitting the model to the experimental data was Markov Chain Monte Carlo (MCMC) sampling, which is a statistical technique that allows us to efficiently explore the parameter space and obtain estimates of the most likely parameter values that account for the observed data.

To evaluate the performance of each parameter combination from the aforementioned methods, log-likelihood maximization was used, as in K rting et al:

$$\log L(model \mid \{n_i\}) = \sum_{i=1}^5 n_i \log p_i + constant. \quad (4)$$

3 Discussion and Results

the causal inference model seems to give logical results that mimic what you would expect in an experiment, to this extent, we analyze this model by examining the relationships between the variables and parameters of the model. in figure 1 the posterior probability of stimuli having a common cause $p(C = 1 \mid x_V, x_A)$ is plotted against x_V while setting $x_A = 0$, thus the x-axis also indicates the spatial disparity between the stimuli percepts. in each panel of the plot, we examine the effect of varying each of the model parameters, namely, (p_{common} , σ_v , σ_a , and σ_p). we can conclude from these plots that:

- increasing the prior of a common cause p_{common} increases the posterior, even for higher disparities.
- increasing the noise in either sensory percepts $\sigma_{v,a}$ increases disparity and thus decreases the posterior.

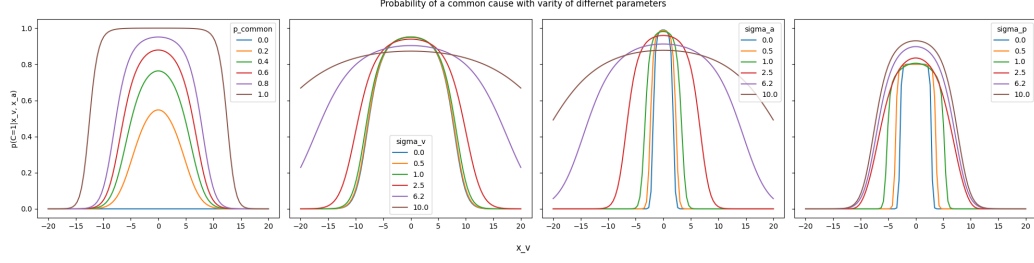


Figure 1: effect of varying model parameters on the posterior probability of stimuli having a common cause $p(C = 1 | x_V, x_A)$ vs visual percept x_V for $x_A = 0$,

- increasing σ_p increases the posterior since it corresponds to a lower bias or prior assumption about the location of the stimuli.

In figure 2 and 3 we examine the relationship of the stimulus position estimates against x_V , in a similar fashion to figure 1.

the plots show that:

- increasing the prior of a common cause p_{common} affects the estimated auditory stimulus position $\hat{s}_a$ such that it makes it follow x_v since vision is the more reliable sense, but not the other way around.
- increasing the noise in either sensory percepts $\sigma_{v,a}$ for $p_{common}=0.8$ makes the estimate arising from the more noisy percept follow the less noisy percept since it is more reliable and there is a high prior probability for a common cause.
- decreasing σ_p corresponds to a stronger bias for expecting stimuli to be in a central location, which is clearly visible as the decrease in line slopes in the fourth panel of Figure 2.

For the next step in our analysis, we generated an experimental dataset and did model fitting using the brute-force search and MCMC sampling as mentioned in the materials and methods section. we now discuss some important findings.

Firstly, we used $N = 100000$ datapoints to generate data for each model

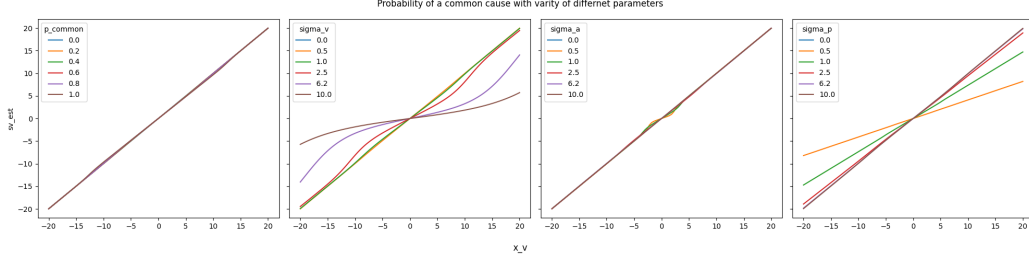


Figure 2: effect of varying model parameters on the estimated visual stimulus position $\hat{s}_v$ vs visual percept x_v for $x_A = 0$,

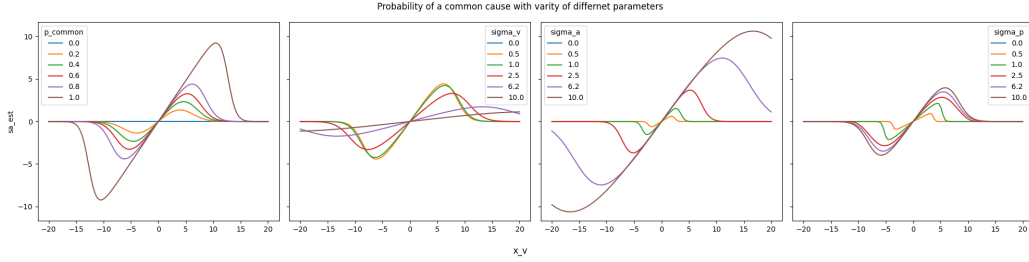


Figure 3: effect of varying model parameters on the estimated auditory stimulus position $\hat{s}_a$ vs visual percept x_v for $x_A = 0$,

being tested, by increasing the amount of data 10-fold, we make sure to achieve stability in p_i in equation 4, i.e., the generated data more accurately represent the model being tested and the calculated log-likelihood values are more robust. Additionally, the log in the aforementioned equation resulted in numerical instability when it evaluates to negative infinity, which means that our log-likelihood function has discontinuities, which is an argument against using gradient-based methods since calculating gradients for such a function is difficult. adding a small value to the log in the equation solves these numerical instabilities.

we note that using the brute-force search method resulted in fit parameters that are similar but still different from the actual parameters used to generate the data, a particular result that we got is: $p_{common}=0.56, \sigma_v=0.001, \sigma_a=4.4, \sigma_p=13.33$, as opposed to the true model parameters which are: $p_{common} = 0.8, \sigma_v = 0.6, \sigma_a = 3.1, \sigma_p = 15$ as mentioned in the parameter initialization section.

Figure 4 shows the marginal likelihood values, and indicates the estimated and true parameters. we note that estimating the actual parameters from

the marginal plots is not possible, since the parameters are not independent.

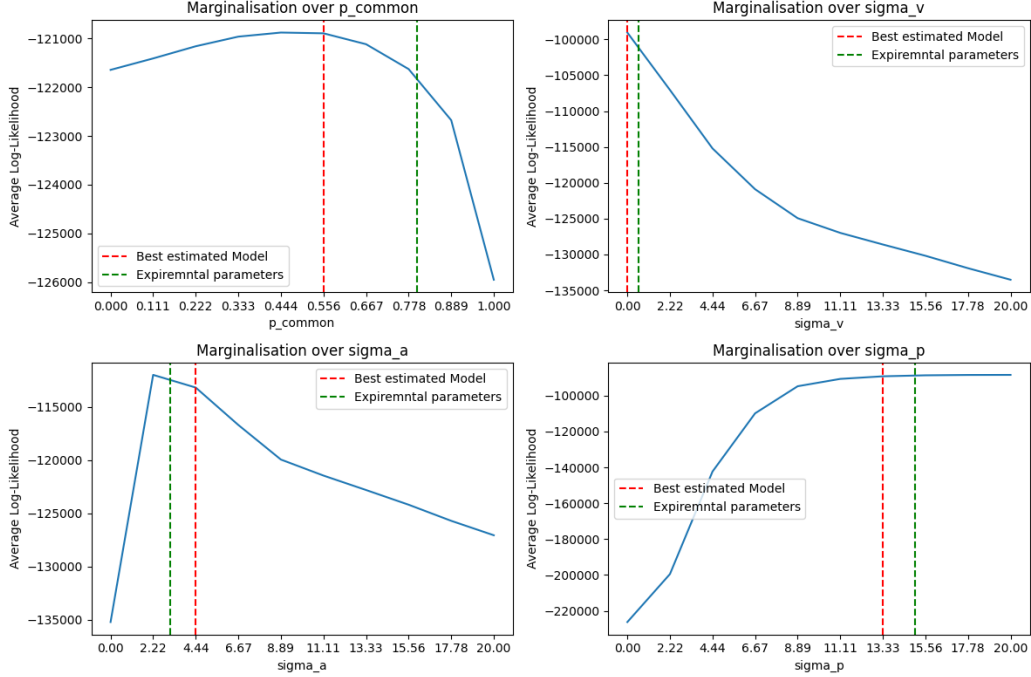


Figure 4: average marginal likelihood values

The MCMC sampling method was more successful in fitting the model, this can be shown in Figure 5. additionally, example estimated parameters obtained from this method are: $p_{\text{common}} = 0.75$, $\sigma_v = 0.56$, $\sigma_a = 3.2$, $\sigma_p = 16.2$.

As a comparison between the two methods, we note that the brute-force parameter space search is a straightforward method, which theoretically can give accurate estimates for any model fit problem no matter how hard it is, given that we use fine steps and examine the whole parameter space, however, this is practically not possible due to the computational power this would require. On the other hand, MCMC is a more involved method that samples the parameter space efficiently and thus is more suitable for high dimensional and large parameter spaces, however, the disadvantage of this method lies in its complexity, that is, the imperative need for hyperparameter tuning for this method to work properly.

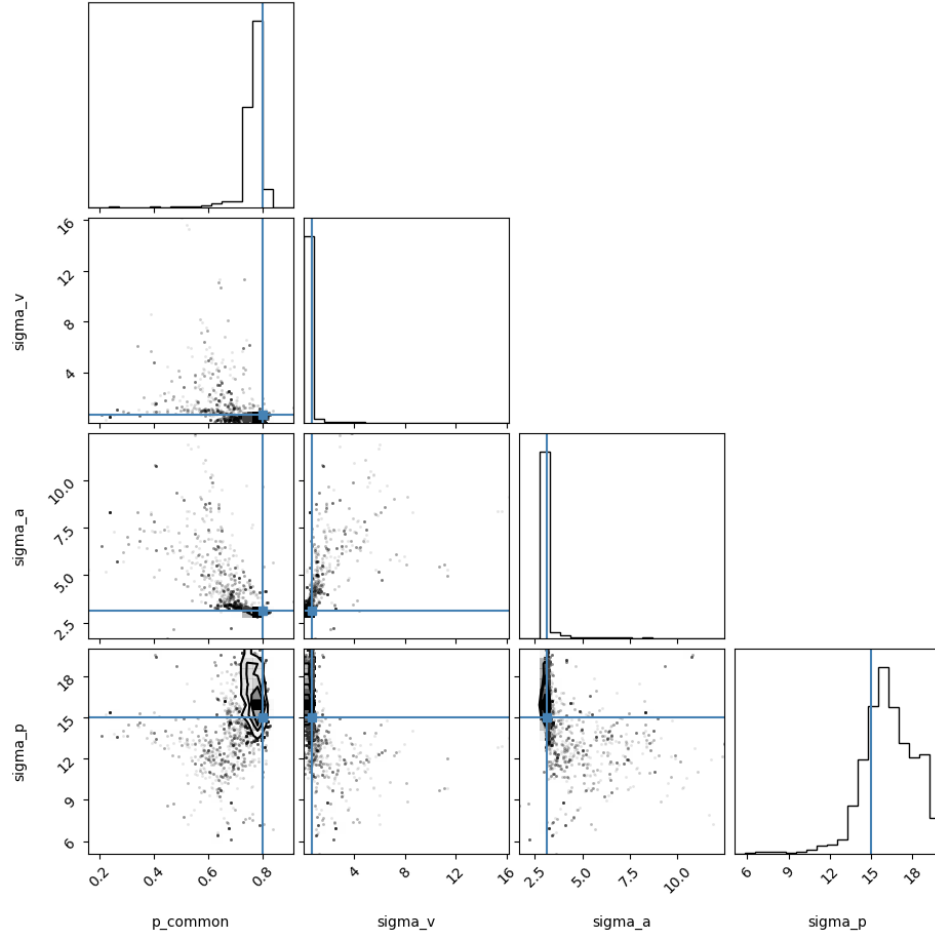


Figure 5: Results of using MCMC sampling for model fitting

Finally, to test the MCMC method in a more heterogenous setting, a single dataset was created by concatenating two datasets generated from two different parameter combinations. Figure 6 shows the results of this experiment.

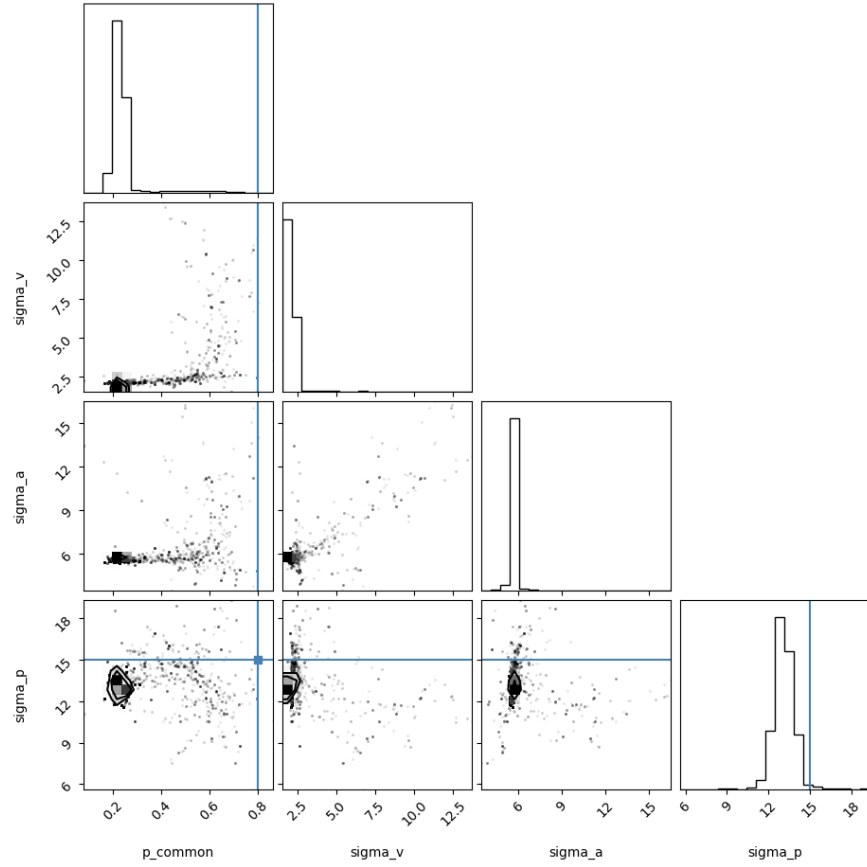


Figure 6: Results of using MCMC sampling for model fitting using a single dataset generated with two sets of parameters.

References

- [1] Koerding, K. P., Beierholm, U., Ma, W. J., Quartz, S., Tenenbaum, J. B., and Shams, L. (2007). Causal inference in multisensory perception. PLoS one, 2(9):e943.